End-to-End Enterprise Credit Risk Architecture

Basel-Aligned Scoring, Deterministic AI Q&A Engine & Autonomous Anomaly Monitoring

Portfolio Grain: 1 row per account_id (14,685 rows)
Stack: Python 3.13 | scikit-learn | Streamlit | LangChain

1. Raw Storage — 5 CSVs

Core Customer
Demographics, income, residence tenure, employment status.
`customers.csv`

Loan Accounts
Exposure, interest rates, tenures, product types, default target.
`loan_accounts.csv`

Child Relational Data
Bureau credit lines, monthly repayments, collections logs.
`3 1:N Child CSVs`

2. Ingestion & ETL — src/data_pipeline.py

Cleaning & Imputation
Categorical standardization (M/F → Male/Female). Skew-safe median cohort imputation for missing values.
`pandas | numpy`

Cartesian Protection
Pre-aggregates 1:N repayment and bureau files before joins, maintaining exact 14,685 account grain.
`GroupBy Aggregations`

3. Predictive Scoring — src/model.py

Champion Scorecard
Balanced Logistic Regression outperforming tree models for regulatory transparency.
`scikit-learn | StandardScaler`

Basel Metrics & Tiers
Computes ROC-AUC (0.6727), KS Stat (26.13%), Gini (0.3455). Segments into Low, Med, High risk tiers.
`Calibrated Probabilities`

4. AI Engines — Dual Engine

AI 1: Anomaly Detector
Evaluates 3 rule algorithms: DPD spikes, peer shortfall outliers (>2.5 SD), and hidden bureau divergence.
`src/anomaly_detector.py`

AI 2: Grounded Q&A
Two-tier architecture: Deterministic Python aggregation executed first; LLM formats synthesized output.
`src/ask_portfolio.py`

5. Analyst Dashboard — app/app.py

Interactive 4-Tab UI

1. **Portfolio View:** Macro KPIs & slice-and-dice.
2. **Account Deep Dive:** Gauge & top drivers.
3. **Ask Portfolio:** Natural language assistant.
4. **Anomaly Queue:** Workflow triage.

`Streamlit | Plotly`

⚠️ AI Feature 1: Transparent Anomaly Detection Pipeline

- **Architecture:** Fully vectorized rules executing during ETL. No opaque black-box scoring.
- **Rule 1 (Sudden Delinquency Jump):** `repay_missed_count == 0` but `repay_max_dpd >= 30`.
- **Rule 2 (Cohort Outlier):** Shortfall > `cohort_mean + 2.5 * std` within identical loan product.
- **Rule 3 (Bureau Divergence):** Clean internal repayment, but `bureau_written_off >= 1` or `bureau_dpd >= 60`.
- **Workflow:** Populates interactive triaging queue with *Confirm* and *Dismiss* audit actions.

💬 AI Feature 2: Provider-Based Grounded LLM Q&A ("Ask the Portfolio")

Anti-Hallucination Design: The LLM is NEVER sent raw tabular records or allowed to execute unverified math. Python calculates the exact metrics first.

Where the LLM is Called & Data Sent:
• **Location:** Triggered in `src/llm_provider.py` via `generate_llm_response()`.
• **Provider Integration:** Configurable via `.env` for **LangChain ChatOpenAI** (supporting custom base_url & model) or direct **OpenAI SDK**.
• **Payload Sent:** User query + *pre-computed factual summary* (e.g., `{"highest_type": "Personal Loan", "rate": "22.4%", "source": ["loan_type", "target_default"]}`).
• **Fallback:** If no API key is provided, routes to **Deterministic Offline Mode** (formats computed facts without API calls).